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DATA ANALYTICS CAPSTONE PROJECT

Workforce Intelligence & Organizational Performance Analysis

Roblox Africa Operations — Ghana Hub

An integrated analysis of **30,004 employees** across HR, finance, health, education and performance systems

TOOLS & TECHNOLOGIES



Microsoft Excel

Data cleaning & exploratory analysis



SQL

Database design, joins & queries



Power BI

Executive dashboards & visualization

Siloed Workforce Data Blocks Strategic Decisions

Five Years of Investment

Roblox Africa operates from its **Ghana regional hub**, spanning engineering, data, customer support, and operations. Leadership has invested heavily in **talent acquisition, employee development, health benefits, and compensation** to fuel rapid growth.

5 years of growing workforce investment without a unified analytical view

6 isolated systems holding people data — HR, finance, health, departmental, employee & department performance

The Core Questions

- ? Are we hiring and placing the right talent in the right departments?
- 🎓 How does education background influence department performance and compensation?
- 💓 Are health and benefits investments translating into productivity?
- ⚠️ Are there hidden risks in workforce distribution, costs, or employee wellbeing?

The Cost of Fragmentation

Without a consolidated view, leadership cannot understand **true workforce costs**, assess education-placement alignment, monitor wellbeing risks, or spot staffing inefficiencies.

The result: decisions around **hiring, promotion, and benefits remain reactive rather than evidence-based**. This project changes that.

Seven Integrated Datasets, One Analytical Foundation

The Data Foundation

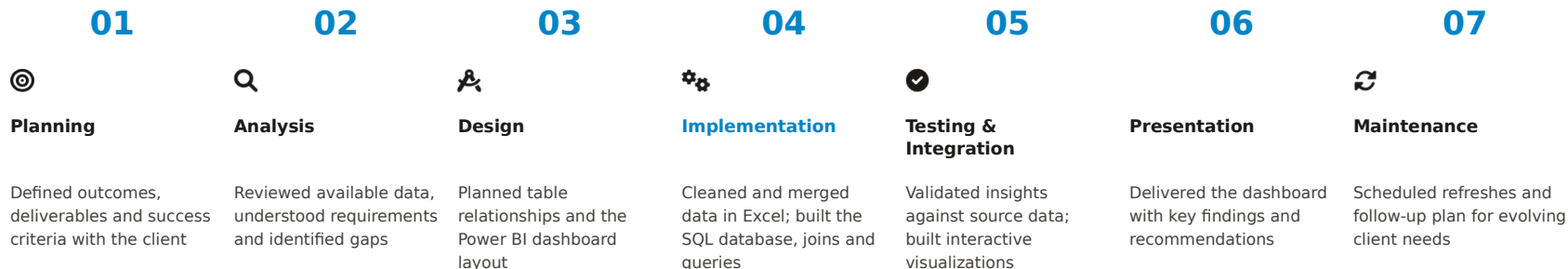
DATASET	RECORDS	ROLE IN THE ANALYSIS
Employee Dataset	30,004	Central dimension table linking all other data
Department Dataset	8	Lookup table mapping department codes to names
Education Dataset	30,001	Human capital quality and academic background
Health Dataset	30,000	Wellbeing indicators and operational risk
Finance Dataset	30,000	Compensation, cost, and value linkage
Employee Performance	150,000	Five-year individual productivity history
Department Performance	40	Eight departments × five years of aggregate targets

Analytical Objectives

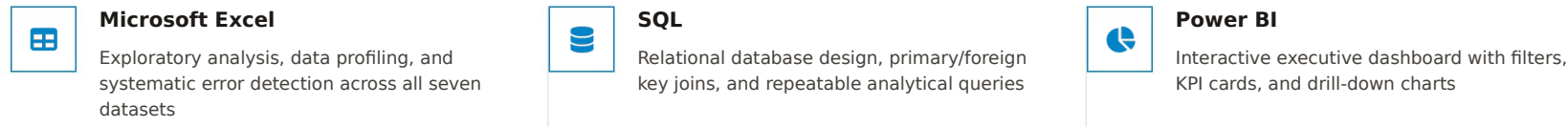
-  **Workforce composition** by department, role, and gender
-  **Education-placement alignment:** degree level and field of study against job placement
-  **Compensation patterns** and cost distribution across the organization
-  **Health-related operational risks** including insurance and leave exposure
-  **Department staffing efficiency** and cost-to-value conversion

Each dataset connects through the **employee_id** as the primary key, forming a single star-schema-style analytical model.

A Full Analytics Lifecycle, Powered by Excel, SQL & Power BI



Where Each Tool Played Its Part



A Unified 30K-Employee View at a Glance

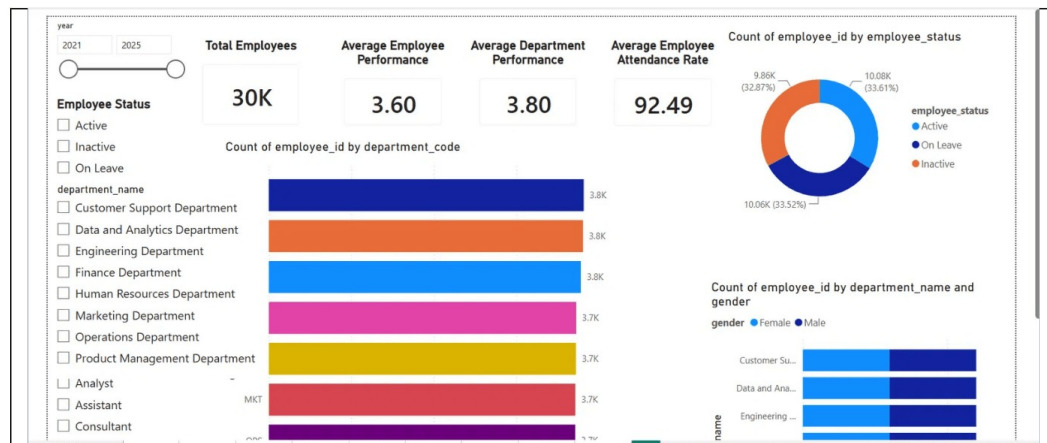


Fig. 1 — Interactive Power BI dashboard: filters for year (2021-2025), status, department, position and gender

30,004

total employees tracked across all departments

3.60

average employee performance score (scale 1-5)

3.80

average department performance score

92.49%

average employee attendance rate

What the Dashboard Reveals

- Employee status splits almost perfectly three ways: **Active 10.08K (33.6%)**, **On Leave 10.06K (33.5%)**, and **Inactive 9.86K (32.9%)** — the unusually high On Leave share is an early signal worth monitoring.
- Departmental staffing is balanced at **~3,700-3,820 employees per department**, led by Customer Support (3,820) — no single department is over- or under-weighted.
- Interactive slicers let leadership drill down by **year, status, department, position, and gender** to explore any segment instantly.

Balanced Departments with Equal Gender Representation

3,699

–
3,820

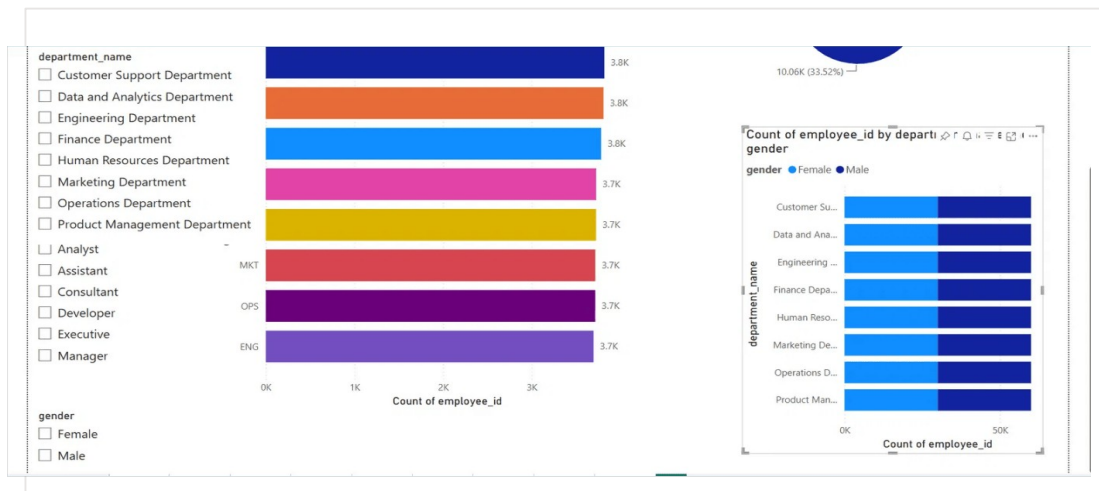
employees per department — remarkably even staffing across all **eight departments**, led by Customer Support (3,820)

5
0
/
5
0
6

15,009 male and 14,987 female employees — a near-perfect gender split, with every department showing a healthy mixed breakdown

roles split evenly at **~4,900-5,100 each**: Executives, Consultants, Analysts, Developers, Assistants, Managers — Executives slightly over-represented at 5,125

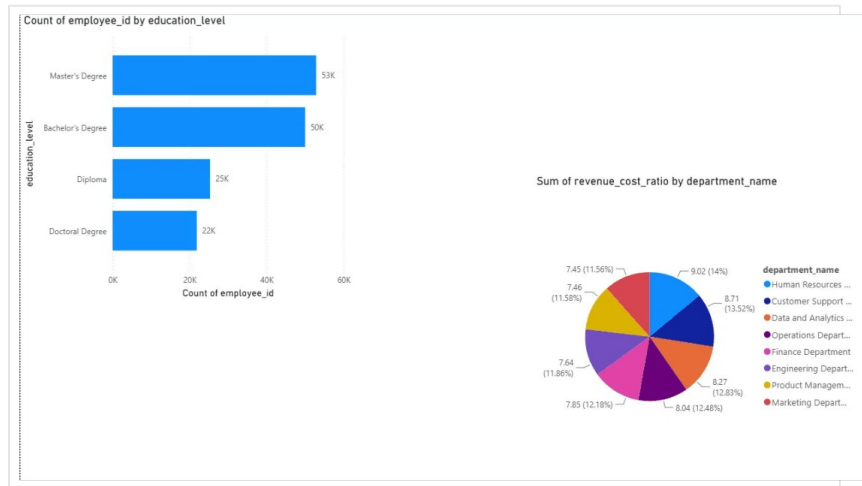
Even distribution across departments and gender signals **equitable hiring and workforce planning** — no single team or demographic is disproportionately burdened.



Source: Power BI executive dashboard — Count of employee_id by department and gender

Every department carries a balanced male/female mix, and headcount variance between the largest (CST) and smallest (ENG) departments is under 4%.

A Highly Educated Workforce Anchored by Business and Tech Degrees



Source: Power BI executive dashboard — Count of employee_id by education_level; sum of revenue_cost_ratio by department

Master's and Bachelor's degrees dominate the workforce, together accounting for 68% of all employees.

Education Level Distribution

10,573 **Master's degrees** — the largest group at 35% of the workforce

9,996 **Bachelor's degrees** — 33%, the backbone of entry and mid-level roles

5,061 **Diplomas** — 17%, mostly technical and support tracks

4,371 **Doctoral degrees** — 15%, a deep specialist bench

83% hold **at least a bachelor's degree** — well above typical industry benchmarks

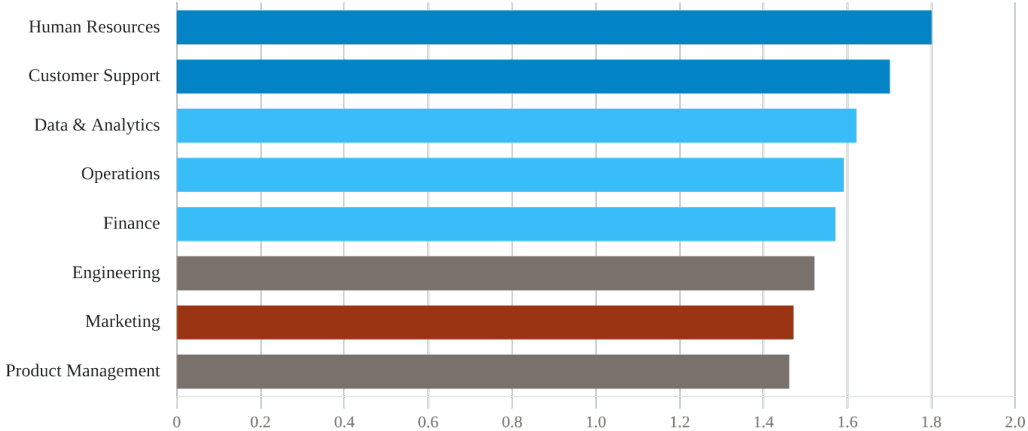
Top Fields of Study

Business Administration leads with **8,554**; Information Technology (**4,326**); Computer Science (**4,318**); Data Analytics (**4,280**); Economics (**4,271**); Engineering (**4,118**).

💡 Implication: the education base aligns well with the department structure — tech roles for IT/CS graduates, finance roles for Economics graduates — while granular education-to-role mapping remains an

Which Departments Convert Cost into Value

Revenue-Cost Ratio by Department (2021-2025)



3.60

org-wide average performance score

92.49%

average attendance rate

49.8%

of employee-years earned bonuses

Source: Department Performance dataset, aggregated 2021-2025; computed with SQL revenue/cost aggregates and Power BI visualization

Average Performance Score (1-5)

DEPARTMENT	SCORE	RATIO
Finance	3.99	1.57
Customer Support	3.98	1.70
Human Resources	3.98	1.80
Operations	3.97	1.59
Data & Analytics	3.78	1.62
Product Management	3.61	1.46
Engineering	3.56	1.52
Marketing	3.50	1.47

🏆 **Finance leads on raw performance (3.99), while HR delivers the highest return per dollar spent (1.80)** — the most efficient department in the organization.

⚠️ **Marketing records the lowest performance score (3.50) and second-lowest efficiency (1.47)** — the clearest candidate for a targeted process and

Compensation, Leave, and Insurance Gaps to Watch



\$312.5M

Compensation is the largest cost line

Average basic salary of **\$8,765 (USD)** plus allowances across 30,000 employees — a major cost line demanding continuous monitoring.



~33%

Leave burden strains capacity

10,057 of 30,004 employees are On Leave at any point — roughly one-third of the workforce — creating persistent backfill pressure on departments.



19,990 / 30,000

Two-thirds lack Active insurance coverage

10,090 policies are Pending and **9,920 are Expired** — only 9,990 employees hold Active coverage, a significant wellbeing and compliance risk.



50.3%

Medical-leave eligibility is concentrated

15,107 employees are medical-leave eligible, concentrating absence risk in specific workforce segments.

THE TAKEAWAY

The health dataset is an **early-warning system** for operational risk.

None of these signals require medical interpretation — they are **workforce operations indicators**: payroll exposure, capacity strain, and coverage gaps that directly affect continuity and compliance.

→ **Priority action:** resolve the 19,990 Pending and Expired insurance policies to close the largest wellbeing and compliance gap identified in this analysis.

From Insights to Action

01



Investigate the On Leave Concentration

Roughly **one-third of employees (33%)** are On Leave at any point — review leave policy and backfill planning to protect capacity.

02

Clear the Insurance Backlog

Renew or resolve the **19,990 Pending and Expired** policies to close a significant wellbeing and compliance risk.

03



Support Marketing & Engineering

Performance scores of **3.50 and 3.56** lag HR, Finance and Customer Support (~3.98) — target training and process improvements.

04



Institutionalize the Data Model

Adopt the **SQL schema and Power BI dashboard** as the standard operating view, with scheduled refreshes and documented maintenance.

05



Extend to Predictive Analytics

Use this foundation to build **attrition forecasting** and workforce planning models.

THE ROAD AHEAD

Each recommendation is rooted in the evidence produced by this project — not intuition. The dashboard is ready to serve as the single, unified operating view for workforce decisions, with a maintenance plan to keep it current as the organization grows.

CLOSING THOUGHT

Integrated, evidence-based workforce analytics turns fragmented HR, finance, and health data into a strategic executive asset.

Power BI Dashboard:

https://app.powerbi.com/links/HjAzZIoOct?ctid=bd697c1b-c481-479c-841e-c618542675c3&pbi_source=linkShare&bookmarkGuid=23f6ca0a-0447-4389-a8b2-bb1540bb9b99